

lasso all concept from scratch to advanced

LASSO Regression: Complete Guide

1. Introduction & Core Concept

What is LASSO?

LASSO (Least Absolute Shrinkage and Selection Operator) is a **regularized regression technique** that:

- Performs both **variable selection** and **regularization**
- Adds L1 penalty to linear regression
- Shrinks some coefficients to **exactly zero**, effectively selecting features

Mathematical Formulation:

text

Minimize: $RSS + \lambda * \sum |\beta_j|$

Where:

RSS = Residual Sum of Squares

λ = Regularization parameter

β_j = Coefficients

2. Mathematical Foundation

Standard Linear Regression

text

Minimize: $\sum (y_i - \beta_0 - \sum \beta_j x_{ij})^2$

LASSO Objective Function

text

$$\text{Minimize: } \sum (y_i - \beta_0 - \sum \beta_j x_{ij})^2 + \lambda \sum |\beta_j|$$

Subject to: $\sum |\beta_j| \leq t$ (budget constraint)

Geometric Interpretation

- **OLS:** Finds β that minimizes RSS (ellipsoids)
- **LASSO:** Constraint region is a **diamond** (L1 norm)
- Solution occurs where ellipsoid touches diamond **at corners**
- Corners = sparse solutions (some coefficients = 0)

3. Key Properties

Sparsity

text

$$\hat{\beta}_{\text{lasso}} = \operatorname{argmin}\{||y - X\beta||^2 + \lambda ||\beta||_1\}$$

- L1 penalty creates **corner solutions**
- Some coefficients exactly zero
- Automatic feature selection

Bias-Variance Tradeoff

- Increases bias but reduces variance
- Optimal λ balances over/under-fitting

Scale Sensitivity

- Features must be **standardized** (mean=0, var=1)
- Penalty treats all coefficients equally only after scaling

4. Solution Methods

Coordinate Descent (Most Common)

For each j in $1 \dots p$:

text

$$\beta_j = S(\sum x_{ij}(y_i - \hat{y}_i^{(-j)}), \lambda) / \sum x_{ij}^2$$

Where S is the soft-thresholding operator:

text

$$S(z, \lambda) = \text{sign}(z)(|z| - \lambda)_+$$

Subgradient Equations

Stationary conditions:

text

$$\begin{aligned} -2\sum x_{ij}(y_i - \hat{y}_i) + \lambda \cdot \text{sign}(\beta_j) &= 0 \text{ if } \beta_j \neq 0 \\ |2\sum x_{ij}(y_i - \hat{y}_i)| &\leq \lambda \text{ if } \beta_j = 0 \end{aligned}$$

LARS Algorithm

Least Angle Regression can compute entire LASSO path efficiently.

5. Tuning Parameter Selection

Cross-Validation

python

```
# Example using scikit-learn
from sklearn.linear_model import LassoCV
```

```
lasso_cv = LassoCV(cv=5, alphas=np.logspace(-4, 0, 100))
lasso_cv.fit(X_train, y_train)
optimal_lambda = lasso_cv.alpha_
```

Information Criteria

- AIC: $-2\log L + 2k$
- BIC: $-2\log L + \log(n)k$ (preferred for selection)

Theoretical Results

- **Consistency:** Under conditions, selects true model
- **Prediction error:** Optimal $\lambda \sim \sqrt{(\log p / n)}$

6. Advanced Topics

Adaptive LASSO

Weights penalties to reduce bias:

text

Minimize: $\sum (y_i - \beta_0 - \sum \beta_j x_{ij})^2 + \lambda \sum w_j |\beta_j|$

Where $w_j = 1/|\beta_{init}|^\gamma$

Group LASSO

Selects groups of variables:

text

Penalty: $\lambda \sum ||\beta_g||_2$

- Useful for categorical variables, interactions

Fused LASSO

Encourages similarity between adjacent coefficients:

text

$$\text{Penalty: } \lambda_1 \sum |\beta_j| + \lambda_2 \sum |\beta_j - \beta_{j-1}|$$

- For ordered features, time series

Elastic Net

Combines L1 and L2 penalties:

text

$$\text{Penalty: } \lambda [\alpha \sum |\beta_j| + (1-\alpha) \sum \beta_j^2]$$

- Handles correlated features better than LASSO

7. Statistical Theory

Oracle Properties

Under appropriate conditions:

1. Identifies correct subset with probability $\rightarrow 1$
2. Estimates have optimal asymptotic distribution

Irrepresentable Condition

For consistent selection, need:

text

$$\|X_{S^c}' X_S (X_S' X_S)^{-1} \text{sign}(\beta_S)\|_\infty < 1$$

Where S = true support

Degrees of Freedom

text

$$df(\lambda) = E[\# \text{ of non-zero coefficients}]$$

- Unbiased estimate for prediction error

8. Computational Aspects

Path Algorithms

Compute solutions for all λ :

python

```
from sklearn.linear_model import lars_path

alphas, _, coefs = lars_path(X, y, method='lasso')
```

Big Data Implementation

python

```
# Stochastic Coordinate Descent
for epoch in range(n_epochs):
    for j in random_permutation(p):
        grad = compute_gradient(j, mini_batch)
         $\beta_j$  = soft_threshold( $\beta_j - \eta \cdot \text{grad}$ ,  $\eta \cdot \lambda$ )
```

Parallelization

- Features can be updated in parallel with synchronization
- Use consensus ADMM for distributed settings

9. Practical Considerations

Preprocessing Checklist

1. Standardize features (center & scale)
2. Handle missing values
3. Check multicollinearity
4. Consider interactions if domain knowledge suggests

Model Diagnostics

python

```
# Check selected features
selected = np.where(coef != 0)[0]
print(f"Selected {len(selected)}/{X.shape[1]} features")

# Check performance
from sklearn.metrics import mean_squared_error
mse = mean_squared_error(y_test, y_pred)
```

Common Pitfalls

1. λ too large: Underfitting, too sparse
2. λ too small: Overfitting, not sparse enough
3. Correlated features: LASSO picks one arbitrarily
4. $p \gg n$: Can select at most n features

10. Extensions & Variations

Sparse Group LASSO

text

Penalty: $\alpha \lambda \sum |\beta_j| + (1-\alpha) \lambda \sum ||\beta_g||_2$

Time-Varying LASSO

For dynamic coefficient models

Quantile LASSO

text

Minimize: $\sum \rho_{\tau}(y_i - x_i' \beta) + \lambda \sum |\beta_j|$

Where ρ_{τ} is quantile loss

Robust LASSO

Use Huber loss instead of squared loss for outliers

11. Implementation Examples

Basic LASSO

python

```
import numpy as np
from sklearn.linear_model import Lasso
from sklearn.preprocessing import StandardScaler

# Scale data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Fit LASSO
lasso = Lasso(alpha=0.1, max_iter=10000)
lasso.fit(X_scaled, y)

# Get results
coef = lasso.coef_
selected = X.columns[coef != 0]
```

Cross-Validation with Early Stopping

python

```

class EarlyStoppingLasso:
    def __init__(self, alpha, tol=1e-4, patience=10):
        self.alpha = alpha
        self.tol = tol
        self.patience = patience

    def fit(self, X, y):
        n_samples, n_features = X.shape
        beta = np.zeros(n_features)
        best_loss = np.inf
        no_improve = 0

        for iteration in range(1000):
            for j in range(n_features):
                # Coordinate descent update
                r = y - X @ beta + X[:, j] * beta[j]
                beta[j] = self.soft_threshold(X[:, j] @ r, self.alpha)

            loss = self.compute_loss(X, y, beta)
            if loss > best_loss - self.tol:
                no_improve += 1
                if no_improve >= self.patience:
                    break
            else:
                best_loss = loss
                no_improve = 0

        self.coef_ = beta
        return self

    def soft_threshold(self, z, gamma):
        return np.sign(z) * max(abs(z) - gamma, 0)

```

12. Recent Advances (2020+)

De-biased LASSO

Reduces bias in coefficient estimates for inference

Variational Inference for LASSO

Bayesian approaches with spike-and-slab priors

Deep LASSO

LASSO penalties in neural networks for interpretability

Federated LASSO

Privacy-preserving distributed LASSO

13. Best Practices Summary

1. **Always standardize** features before applying LASSO
2. Use **cross-validation** to select λ
3. Check **stability** of selected features via bootstrap
4. Consider **Elastic Net** if features are correlated
5. Use **warm starts** for path computation
6. For inference after selection, use **double LASSO** or de-biased methods

14. Applications

- **Genomics:** Select relevant genes from thousands
- **Finance:** Portfolio optimization with sparse holdings
- **Image Processing:** Sparse representation
- **Natural Language Processing:** Feature selection in text
- **Economics:** High-dimensional causal inference

Further Resources

Books

1. *Statistical Learning with Sparsity* - Hastie, Tibshirani, Wainwright
2. *High-Dimensional Statistics* - Wainwright

Key Papers

1. Tibshirani (1996) - Original LASSO paper
2. Zou & Hastie (2005) - Elastic Net
3. Meinshausen & Bühlmann (2006) - Stability selection
4. Candès & Tao (2007) - Dantzig selector

Software

- **Python:** scikit-learn, glmnet-python, statsmodels
- **R:** glmnet, lars, ncvreg
- **Julia:** Lasso.jl, GLMNet.jl

This comprehensive guide covers LASSO from fundamentals to advanced topics. The method remains a cornerstone of high-dimensional statistics and machine learning due to its elegant combination of prediction accuracy and interpretability.

explain mathematics formula

LASSO Mathematics: Complete Formula Breakdown

1. Standard Linear Regression

Ordinary Least Squares (OLS)

Objective:

text

$$\text{minimize: } \text{RSS}(\beta) = \sum_{i=1}^n (y_i - x_i^T \beta)^2$$

Where:

- y_i : i-th observation of dependent variable
- x_i : p-dimensional vector of features for i-th observation
- β : p-dimensional coefficient vector
- RSS : Residual Sum of Squares

Matrix Notation:

text

minimize: $\|y - X\beta\|_2^2$

Where:

- $y \in \mathbb{R}^n$: response vector
- $X \in \mathbb{R}^{n \times p}$: design matrix
- $\|\cdot\|_2$: L2 norm (Euclidean norm)

Solution (normal equations):

text

$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$

2. LASSO Formulation

Constrained Form

text

minimize: $RSS(\beta) = \sum (y_i - x_i^T \beta)^2$

subject to: $\|\beta\|_1 \leq t$

Where:

- $\|\beta\|_1 = \sum |\beta_j|$: L1 norm (sum of absolute values)

- $t \geq 0$: tuning parameter (budget constraint)

Lagrangian Form (Common Form)

text

$$\text{minimize: } L(\beta) = (1/2n) \sum (y_i - x_i^T \beta)^2 + \lambda \|\beta\|_1$$

Where:

- $\lambda \geq 0$: regularization parameter
- $1/(2n)$ factor often included for scaling

Complete Matrix Form:

text

$$\text{minimize: } (1/(2n)) \|y - X\beta\|_2^2 + \lambda \|\beta\|_1$$

Relationship between λ and t

For every $\lambda \geq 0$, $\exists t \geq 0$ such that:

text

$$\text{argmin}_{\beta} (1/(2n)) \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 = \text{argmin}_{\beta} \|y - X\beta\|_2^2 \text{ s.t. } \|\beta\|_1 \leq t$$

3. The Soft-Thresholding Operator

Definition

text

$$S(z, \lambda) = \text{sign}(z) \cdot \max(|z| - \lambda, 0)$$

Piecewise Representation

text

$$S(z, \lambda) = \begin{cases} z - \lambda & \text{if } z > \lambda \\ 0 & \text{if } |z| \leq \lambda \\ z + \lambda & \text{if } z < -\lambda \end{cases}$$

Geometric Interpretation

For scalar LASSO problem:

text

$$\text{minimize: } (1/2)(z - \beta)^2 + \lambda|\beta|$$

Solution: $\hat{\beta} = S(z, \lambda)$

4. Coordinate Descent Update Rule

For Standardized Features (mean=0, variance=1)

Partial residual:

text

$$r_i^{(j)} = y_i - \sum_{k \neq j} \beta_k x_{ik}$$

Univariate OLS estimate:

text

$$\beta_{j_OLS} = (1/n) \sum_{i=1} x_{ij} r_i^{(j)}$$

LASSO Update:

text

$$\beta_{j_new} = S(\beta_{j_OLS}, \lambda)$$

Complete Algorithm

For each $j = 1, \dots, p$:

text

$$\beta_j \leftarrow S\left(\frac{1}{n} \sum_{i=1}^n x_{ij}(y_i - \sum_{k \neq j} \beta_k x_{ik}), \lambda\right)$$

Matrix form update:

text

$$\beta_j \leftarrow S\left(X_j^T(y - X\beta + X_j\beta_j)/n, \lambda\right)$$

5. Karush-Kuhn-Tucker (KKT) Conditions

Subgradient of Absolute Value

text

$$\partial|\beta_j| = \begin{cases} \text{sign}(\beta_j) & \text{if } \beta_j \neq 0 \\ [-1, 1] & \text{if } \beta_j = 0 \end{cases}$$

KKT Conditions for LASSO

For optimality, $\exists g_j \in \partial|\beta_j|$ such that:

text

$$(1/n)X_j^T(X\beta - y) + \lambda g_j = 0 \quad \text{for all } j$$

Component-wise conditions:

1. If $\beta_j \neq 0$:

text

$$(1/n)X_j^T(y - X\beta) = \lambda \cdot \text{sign}(\beta_j)$$

2. If $\beta_j = 0$:

text

$$|(1/n)X_j^T(y - X\beta)| \leq \lambda$$

Stationary Conditions

text

$$|X_j^T(y - X\beta)| \leq n\lambda \quad \text{for all } j$$

$$X_j^T(y - X\beta) = n\lambda \cdot \text{sign}(\beta_j) \quad \text{for } j \text{ where } \beta_j \neq 0$$

6. Degrees of Freedom Formula

Stein's Formula for LASSO

text

$$df(\lambda) = E[|A_\lambda|]$$

Where $A_\lambda = \{j: \beta_j(\lambda) \neq 0\}$ is the active set

More precise (for orthogonal X):

text

$$df(\lambda) = \sum_{j=1} P(|\beta_{j_OLS}| > \lambda)$$

General Case (Approximate)

text

$$df(\lambda) \approx E[\text{tr}(X_A(X_A^T X_A)^{-1} X_A^T)]$$

Where X_A contains columns in active set

7. Pathwise Solution Properties

Piecewise Linear Solution Path

text

$\hat{\beta}(\lambda)$ is piecewise linear in λ

Breakpoints occur when:

text

$$\lambda_k = |X_j^T(y - X\hat{\beta}(\lambda_k))|/n \quad \text{for some } j$$

LARS Algorithm Insight

At each step, the coefficient path follows:

text

$$\hat{\beta}(\lambda) = \hat{\beta}(\lambda_k) + (\lambda_k - \lambda)\gamma \quad \text{for } \lambda \in [\lambda_{k+1}, \lambda_k]$$

Where γ is the LARS direction vector

8. Prediction Error Formulas

MSE Decomposition

text

$$\text{MSE}(\lambda) = \text{Bias}^2(\lambda) + \text{Variance}(\lambda) + \sigma^2$$

Expected Prediction Error

text

$$\text{EPE}(\lambda) = \sigma^2 + (1/n) ||X(\beta(\lambda) - \beta^*)||_2^2 + (\sigma^2/n)df(\lambda)$$

Where β^* is true coefficient vector

Optimal λ Rate

Under sparsity assumption (s non-zero coefficients):

text

$$\lambda^* \sim \sigma \sqrt{(\log p / n)}$$

Gives near-oracle performance

9. Consistency Results

Prediction Consistency

text

$$(1/n) ||X(\beta(\lambda) - \beta^*)||_2^2 \rightarrow 0 \text{ in probability}$$

When $\lambda \rightarrow 0$ and $n\lambda \rightarrow \infty$

Selection Consistency (Irrepresentable Condition)

If:

text

$$||X_{S^c}^T X_S (X_S^T X_S)^{-1} \text{sign}(\beta_S^*)||_\infty < 1$$

And $\lambda \asymp \sqrt{(\log p / n)}$, then:

text

$$P(\{j: \hat{\beta}_j(\lambda) \neq 0\} = S) \rightarrow 1$$

Where S = true support set

10. Elastic Net Formulation

Combined Penalty

text

$$\text{minimize: } (1/(2n)) \|y - X\beta\|_2^2 + \lambda [\alpha \|\beta\|_1 + (1-\alpha)/2 \|\beta\|_2^2]$$

Update Rule for Elastic Net

text

$$\beta_{j_new} = S(\beta_{j_OLS}, \lambda\alpha) / (1 + \lambda(1-\alpha))$$

11. Adaptive LASSO

Weighted Penalty

text

$$\text{minimize: } (1/(2n)) \|y - X\beta\|_2^2 + \lambda \sum w_j |\beta_j|$$

Weights Definition

Typically:

text

$$w_j = 1/|\beta_{j_init}|^\gamma$$

Where $\gamma > 0$, often $\gamma = 1$

Oracle Property

Under reasonable conditions:

- Identifies correct model with probability $\rightarrow 1$
- Has optimal convergence rate

12. Group LASSO

For Groups $g \in G$

text

$$\underset{g \in G}{\text{minimize:}} \quad (1/(2n)) \left\| y - \sum_{g \in G} X_g \beta_g \right\|_2^2 + \lambda \sum_{g \in G} \sqrt{p_g} \left\| \beta_g \right\|_2$$

Where p_g = size of group g

Block Soft-Thresholding

Solution for group g :

text

$$\hat{\beta}_g = (1 - \lambda \sqrt{p_g} / \left\| z_g \right\|_2)_+ z_g$$

Where $z_g = (X_g^T X_g)^{-1} X_g^T r_g$

13. Matrix Form Derivation

Gradient of RSS

text

$$\nabla \text{RSS}(\beta) = -2X^T(y - X\beta)$$

Subgradient of Full Objective

text

$$\nabla L(\beta) = -X^T(y - X\beta)/n + \lambda \cdot g$$

Where $g = [g_1, \dots, g_p]^T$ with $g_j \in \partial|\beta_j|$

Normalization Consideration

If X columns have norm $\sqrt{n}$ (common standardization):

text

$$X_j^T X_j = n \quad \text{for all } j$$

Then update simplifies to:

text

$$\beta_{j_new} = S(\beta_{j_old} + X_j^T(y - X\beta)/n, \lambda)$$

14. Bayesian Interpretation

Laplace Prior

text

$$p(\beta|\sigma^2) \propto \prod_{j=1} \exp(-\lambda|\beta_j|/\sigma^2)$$

Posterior Mode

text

$$\operatorname{argmax}_{\beta} \log p(\beta|y, X) = \operatorname{argmin}_{\beta} (1/(2\sigma^2)) \|y - X\beta\|_2^2 + \lambda \|\beta\|_1$$

Thus, LASSO = MAP estimate with Laplace prior

Scale Parameter

text

$$\lambda = \sqrt{(2\sigma^2)/\tau}$$

Where τ is scale parameter of Laplace distribution

15. Duality Theory

Primal Problem

text

$$\text{minimize: } (1/2) \|y - X\beta\|_2^2 + \lambda \|\beta\|_1$$

Dual Problem

text

$$\begin{aligned} \text{maximize: } & (1/2) \|y\|_2^2 - (1/2) \|y - X\theta/\lambda\|_2^2 \\ \text{subject to: } & \|X^T\theta\|_\infty \leq \lambda, \|\theta\|_\infty \leq 1 \end{aligned}$$

Dual Variables

Optimality condition:

text

$$\hat{\theta} = y - X\hat{\beta}$$

16. Numerical Optimization Details

Proximal Gradient Update

text

$$\beta^{(k+1)} = \text{prox}_{\{\eta\lambda \|\cdot\|_1\}}[\beta^{(k)} - \eta X^T(X\beta^{(k)} - y)]$$

Where:

- η : step size ($0 < \eta < 1/\|X^T X\|$)
- prox : proximal operator

Proximal operator for L1 norm:

text

$$\text{prox}_{\{\eta\lambda \|\cdot\|_1\}}(z) = S(z, \eta\lambda)$$

Convergence Rate

- Coordinate descent: linear convergence
- Proximal gradient: $O(1/k)$ rate

Key Formulas Summary

Concept	Formula
Objective	$\min (1/(2n))\ y-X\beta\ _2^2 + \lambda\ \beta\ _1$
Soft-threshold	$S(z,\lambda) = \text{sign}(z)\max(z -\lambda,0)$
Coordinate Update	$\beta_j \leftarrow S(\beta_{j_OLS}, \lambda)$
KKT Condition	$ X_j^T(y-X\beta) \leq n\lambda$
Active Set	$A = \{j: X_j^T(y-X\beta) = n\lambda\}$
Degrees Freedom	$df(\lambda) = E[A_\lambda]$
Optimal λ Rate	$\lambda^* \sim \sigma\sqrt{(\log p / n)}$

Concept	Formula
Irrepresentable	$ X_S^c{}^TX_S(X_S^TX_S)^{-1}s_S _\infty < 1$

These mathematical formulations provide the complete theoretical foundation for understanding, implementing, and extending LASSO regression in practice.

explain every mathematics formula with name

LASSO Mathematics: Complete Formula Dictionary

1. BASIC REGRESSION FORMULAS

1.1 Ordinary Least Squares (OLS) Objective

Name: Residual Sum of Squares Minimization

text

$$RSS(\beta) = \sum_{i=1} (y_i - \beta_0 - \sum_{j=1} \beta_j x_{ij})^2$$

Variables:

- RSS : Residual Sum of Squares
- y_i : i-th response value
- β_0 : Intercept coefficient
- β_j : j-th slope coefficient
- x_{ij} : j-th feature for i-th observation
- n : Number of observations
- p : Number of features

1.2 Matrix Form of OLS

Name: Quadratic Loss in Matrix Notation

text

$$\text{minimize: } L(\beta) = (y - X\beta)^T(y - X\beta)$$

Variables:

- $y \in \mathbb{R}^n$: Response vector
- $X \in \mathbb{R}^{n \times p}$: Design matrix (with column of 1's for intercept)
- $\beta \in \mathbb{R}^p$: Coefficient vector

1.3 Normal Equations

Name: OLS Closed-Form Solution

text

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$$

Derivation: $\nabla L(\beta) = -2X^T(y - X\beta) = 0 \Rightarrow X^T X \beta = X^T y$

2. LASSO CORE FORMULAS

2.1 Constrained Formulation (Original)

Name: Budget-Constrained L1 Regularization

text

minimize: $RSS(\beta)$

subject to: $\sum_{j=1} |\beta_j| \leq t$

Variables:

- $t \geq 0$: L1-norm budget (tuning parameter)
- $|\beta_j|$: Absolute value of j-th coefficient

2.2 Lagrangian Form (Common)

Name: Penalized Least Squares with L1 Penalty

text

$$\underset{\beta}{\text{minimize:}} \quad L(\beta) = \frac{1}{2n} \sum_{i=1} (y_i - x_i^T \beta)^2 + \lambda \sum_{j=1} |\beta_j|$$

Variables:

- $\lambda \geq 0$: Regularization parameter (Lagrange multiplier)
- $1/(2n)$: Scaling factor for mathematical convenience

2.3 Complete Matrix Form

Name: LASSO Objective in Matrix Notation

text

$$\underset{\beta}{\text{minimize:}} \quad \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1$$

Norms:

- $\|v\|_2^2 = \sum v_i^2$: Squared L2 norm (Euclidean)
- $\|\beta\|_1 = \sum |\beta_j|$: L1 norm (Manhattan/Taxicab)

3. OPTIMIZATION FORMULAS

3.1 Subgradient of Absolute Value

Name: Subdifferential of $|x|$

text

$$\partial |x| = \begin{cases} \text{sign}(x) & \text{if } x \neq 0 \\ [-1, 1] & \text{if } x = 0 \end{cases}$$

Where:

- $\text{sign}(x) = 1$ if $x > 0$, -1 if $x < 0$
- $[-1, 1]$: Set of all values between -1 and 1

3.2 Karush-Kuhn-Tucker (KKT) Conditions**Name:** First-Order Optimality Conditions

text

$$(1/n)X_j^T(X\hat{\beta} - y) + \lambda g_j = 0, \forall j$$

With:

- $g_j \in \partial|\beta_j|$: Subgradient of absolute value
- x_j : j -th column of design matrix

3.3 Stationary Conditions**Name:** Component-wise Optimality Criteria

text

For active coefficients ($\beta_j \neq 0$): $X_j^T(y - X\beta) = n\lambda \cdot \text{sign}(\beta_j)$

For zero coefficients ($\beta_j = 0$): $|X_j^T(y - X\beta)| \leq n\lambda$

3.4 Soft-Thresholding Operator**Name:** Proximal Operator for L1 Norm

text

$$S(z, \lambda) = \text{sign}(z) \cdot \max(|z| - \lambda, 0)$$

Piecewise Definition:

text

$$S(z, \lambda) = \begin{cases} z - \lambda & \text{if } z > \lambda \\ 0 & \text{if } |z| \leq \lambda \\ z + \lambda & \text{if } z < -\lambda \end{cases}$$

3.5 Coordinate Descent Update

Name: Cyclic Coordinate-wise Minimization

text

$$\beta_j^{(k+1)} = S(\beta_j^{(k)} + (1/n)X_j^T(y - X\beta^{(k)}), \lambda)$$

Step-by-step:

1. Compute partial residual: $r = y - \sum_{k \neq j} X_k \beta_k^{(k)}$
2. Compute univariate OLS: $z = \beta_j^{(k)} + (1/n)X_j^T r$
3. Apply soft-threshold: $\beta_j^{(k+1)} = S(z, \lambda)$

4. GEOMETRIC INTERPRETATION FORMULAS

4.1 Constraint Geometry

Name: L1 Ball (Diamond) in p-dimensions

text

$$\{\beta \in \mathbb{R}^p : \|\beta\|_1 = \sum |\beta_j| \leq t\}$$

Properties:

- Vertices at: $(\pm t, 0, \dots, 0)$, $(0, \pm t, \dots, 0)$, etc.
- Contains 2^p vertices in $\mathbb{R}^p$
- Intersection with axes creates sparsity

4.2 Solution Geometry

Name: Ellipsoid-Diamond Intersection

Solution occurs where:

text

Contours of $RSS(\beta) = \text{constant}$

Touch the constraint set:

text

$$\{\beta : \|\beta\|_1 \leq \tau\}$$

At a vertex $\Rightarrow$ some $\beta_j = 0$

5. STATISTICAL PROPERTIES FORMULAS

5.1 Bias-Variance Decomposition

Name: Expected Prediction Error

text

$$\begin{aligned} \text{MSE}(\hat{\beta}_{\lambda}) &= E[\|X(\hat{\beta}_{\lambda} - \beta^*)\|_2^2]/n \\ &= \text{Bias}^2(\hat{\beta}_{\lambda}) + \text{Variance}(\hat{\beta}_{\lambda}) \end{aligned}$$

Where:

- β^* : True coefficient vector
- $\text{Bias}(\hat{\beta}_{\lambda}) = E[\hat{\beta}_{\lambda}] - \beta^*$
- $\text{Variance}(\hat{\beta}_{\lambda}) = E[\|\hat{\beta}_{\lambda} - E[\hat{\beta}_{\lambda}]\|_2^2]$

5.2 Degrees of Freedom

Name: Stein's Unbiased Risk Estimate (SURE)

text

$$\begin{aligned} \text{df}(\lambda) &= E[\text{div}(\hat{y})] \\ &\approx E[|A(\lambda)|] \quad (\text{for orthogonal } X) \end{aligned}$$

Where:

- $\hat{y} = X\hat{\beta}(\lambda)$: Fitted values
- div : Divergence operator
- $A(\lambda) = \{j : \hat{\beta}_j(\lambda) \neq 0\}$: Active set

5.3 Risk Inflation

Name: Price for Not Knowing Support

text

$$R(\lambda) = \sigma^2(1 + (|A| \log p)/n)$$

- σ^2 : Error variance
- $|A|$: Size of selected set

6. CONSISTENCY & SELECTION FORMULAS

6.1 Irrepresentable Condition

Name: Strong Irrepresentable Condition

text

$$\|X_{S^c}^T X_S (X_S^T X_S)^{-1} \text{sign}(\beta_S)\|_\infty < 1$$

Variables:

- $S = \{j : \beta_j^* \neq 0\}$: True support
- S^c : Complement of S

- $x_{\cdot S}$: Columns of X indexed by S
- $\text{sign}(\beta_{\cdot S})$: Vector of signs of true non-zero coefficients

6.2 Sure Screening Property

Name: Probability of Including True Variables

text

$$P(S \subseteq \hat{S}) \rightarrow 1 \text{ as } n \rightarrow \infty$$

Where:

- $\hat{S} = \{j : \hat{\beta}_j(\lambda) \neq 0\}$: Estimated support
- Requires: $\min |\beta_j^*| \geq C\sqrt{\log p/n}$

6.3 Optimal λ Rate

Name: Tuning Parameter Scaling

text

$$\lambda^* = C\sigma\sqrt{2 \log p/n}$$

Where:

- $c > 0$: Constant (often $C = 1$)
- σ : Standard deviation of errors

7. EXTENSION FORMULAS

7.1 Elastic Net

Name: Combined L1 + L2 Regularization

text

$$\text{minimize: } (1/2n) ||y - X\beta||_2^2 + \lambda[\alpha ||\beta||_1 + (1-\alpha)/2 ||\beta||_2^2]$$

Variables:

- $\alpha \in [0, 1]$: Mixing parameter
- $\alpha = 1$: Pure LASSO
- $\alpha = 0$: Ridge regression

7.2 Adaptive LASSO

Name: Weighted L1 Penalty

text

$$\text{minimize: } (1/2n) ||y - X\beta||_2^2 + \lambda \sum_{j=1} w_j |\beta_j|$$

Weights:

text

$$w_j = 1/|\beta_{j_init}|^\gamma$$

Where:

- β_{init} : Initial estimate (e.g., OLS or Ridge)
- $\gamma > 0$: Power parameter ($\gamma = 1$ common)

7.3 Group LASSO

Name: L2 Norm over Groups

text

$$\text{minimize: } (1/2n) ||y - \sum_{g \in G} X_g \beta_g||_2^2 + \lambda \sum_{g \in G} \sqrt{p_g} ||\beta_g||_2$$

Where:

- G : Set of groups
- p_g : Size of group g
- $\|\beta_g\|_2 = \sqrt{\sum_{j \in g} \beta_j^2}$: L2 norm within group $j \in g$

7.4 Fused LASSO

Name: Pairwise Differences Penalty

text

$$\text{minimize: } (1/2n) \|y - X\beta\|_2^2 + \lambda_1 \sum_{j=1} |\beta_j| + \lambda_2 \sum_{j=2} |\beta_j - \beta_{j-1}|$$

Application: Ordered features, time series

8. COMPUTATIONAL COMPLEXITY FORMULAS

8.1 Convergence Rate

Name: Linear Convergence of Coordinate Descent

text

$$\|\beta^{(k)} - \beta^*\|_2 \leq C\rho^k$$

Where:

- $\rho \in (0, 1)$: Convergence rate
- $C > 0$: Constant depending on problem
- Typically $\rho \approx 0.5$ to 0.9

8.2 Computational Cost

Name: Per-Iteration Complexity

text

$O(np)$ operations per iteration

Breakdown:

- Residual update: $O(n)$
- Inner product: $O(n)$ per coefficient
- Soft-threshold: $O(1)$ per coefficient

8.3 Path Algorithm Complexity

Name: LARS/LASSO Path Complexity

text

$O(\min(n^2p, np^2))$ operations

Cases:

- $p > n$: $O(n^2p)$
- $n > p$: $O(np^2)$

9. BAYESIAN INTERPRETATION FORMULAS

9.1 Laplace Prior

Name: Double Exponential Distribution

text

$$p(\beta_j | \tau) = (1/(2\tau)) \exp(-|\beta_j|/\tau)$$

Parameters:

- $\tau > 0$: Scale parameter

- Mean = 0, Variance = $2\tau^2$

9.2 Posterior Mode

Name: Maximum A Posteriori (MAP) Estimate

text

$$\hat{\beta}_{\text{MAP}} = \operatorname{argmax} \log p(\beta|y, X)$$

With:

text

$$\log p(\beta|y, X) \propto -(1/(2\sigma^2))||y - X\beta||_2^2 - (1/\tau)\sum|\beta_j|$$

Equivalence: LASSO with $\lambda = \sigma^2/\tau$

9.3 Full Posterior

Name: Bayesian LASSO

text

$$p(\beta|y, X) \propto \exp(-(1/(2\sigma^2))||y - X\beta||_2^2 - (1/\tau)\sum|\beta_j|)$$

Computation: Requires MCMC or variational inference

10. PREDICTION & INFERENCE FORMULAS

10.1 Prediction Interval

Name: Approximate Prediction Variance

text

$$\operatorname{Var}(\hat{y}_{\text{new}}) \approx \sigma^2[x_{\text{new}}^T(X_A^T X_A)^{-1}x_{\text{new}} + 1]$$

Where:

- x_A : Active set design matrix
- x_{new} : New observation
- Requires: σ^2 estimated from residuals

10.2 De-biased LASSO**Name:** Bias-Corrected Estimator

text

$$\hat{\beta}_{\text{debiased}} = \hat{\beta}_{\text{lasso}} + \Theta X^T (y - X\hat{\beta}_{\text{lasso}})/n$$

Where:

- Θ : Approximate inverse of $\hat{\Sigma} = X^T X/n$
- Constructed via nodewise regression

10.3 Post-Selection Inference**Name:** Conditional Inference

text

$$P(\beta_j = 0 \mid \hat{S} = S) \text{ follows truncated normal}$$

Conditional distribution:

text

$$(X_S^T y \mid \{\text{sign}(\hat{\beta}_S) = s_S, A = S\}) \sim \text{truncated normal}$$

11. SPECIAL CASE FORMULAS**11.1 Orthogonal Design Matrix**

Name: Simplified Solution

text

$$\hat{\beta}_j = S(\hat{\beta}_{j_OLS}, \lambda)$$

When: $X^T X = I$ (orthonormal columns)

11.2 $p > n$ Case

Name: High-Dimensional Regime

Maximum selected variables:

text

$$|A| \leq \min(n, p)$$

Note: LASSO can select at most n variables when $p > n$

11.3 Correlation Structure

Name: Grouping Effect (Elastic Net)

For highly correlated x_j and x_k :

text

$$|\hat{\beta}_j - \hat{\beta}_k| \leq C/\lambda$$

Where C depends on correlation

12. MODEL SELECTION FORMULAS

12.1 Cross-Validation Score

Name: K-Fold CV Error

text

$$CV(\lambda) = (1/K) \sum_{k=1}^K \sum_{i \in \text{fold } k} (y_i - x_i^T \beta^{(-k)}(\lambda))^2$$

Where:

- $\beta^{(-k)}(\lambda)$: LASSO estimate trained without fold k

12.2 Information Criteria

Name: AIC and BIC for LASSO

text

$$\begin{aligned} AIC(\lambda) &= n \log(RSS(\lambda)/n) + 2|A(\lambda)| \\ BIC(\lambda) &= n \log(RSS(\lambda)/n) + \log(n)|A(\lambda)| \end{aligned}$$

Note: These are approximations since LASSO isn't MLE

12.3 Stability Selection

Name: Probability of Selection

text

$$\Pi_j(\lambda) = P(j \in \hat{S}(\lambda, I)) \text{ over random subsets } I$$

Threshold: Select variables with $\Pi_j(\lambda) > \pi_{thr}$ (e.g., 0.6)

QUICK REFERENCE TABLE

Formula Name	Mathematical Expression	Key Variables
OLS	$\min y - X\beta _2^2$	$\beta \in \mathbb{R}^p, X \in \mathbb{R}^{n \times p}$
LASSO Objective	$\min (1/2n) y-X\beta _2^2 + \lambda \beta _1$	$\lambda \geq 0$ tuning parameter

Formula Name	Mathematical Expression	Key Variables
Soft-Threshold	$S(z, \lambda) = \text{sign}(z) \max(z - \lambda, 0)$	z : input, λ : threshold
KKT Conditions	$ X_j^T(y - X\beta) \leq n\lambda \quad \forall j$	Equality for active vars
Coordinate Update	$\beta_j \leftarrow S(\beta_j + X_j^T r / n, \lambda)$	$r = y - X\beta$ residuals
Irrepresentable	$\ X_{S^c}^T X_S (X_S^T X_S)^{-1} X_S^T X_{S^c}\ _\infty < 1$	S : true support